

Q How advanced is an embryo? Which parts of my baby's body develop first?

Even in the very earliest days, your baby is already developing some primitive, functional versions of the fully fledged vital organs she'll have when she's born. As a mass of cells, the embryo is not yet akin to how we might picture a developing baby, but it is brimming with potential.

During the first 10 weeks of pregnancy, your ball of potential undergoes a process called organogenesis—the creation of your baby's vital organs. Development of the organs is far more important than your embryo's growth, which is why she grows very little in length during the first 12 weeks compared with the second and third trimesters of pregnancy. All her cellular energy is diverted into creating her organs. This requires a process of rapid cell division

and the subdivision of groups of cells into their different functions: it's called differentiation.

By five weeks, your baby's thyroid, renal, adrenal, and gonadal organs (glands of her endocrine, or hormonal, system) are present in a primitive form. Cells that will form her brain, kidneys, heart, and nerves have appeared. She has a tube that will become her gastrointestinal tract. Her lungs and stomach have divided from a single tube within the embryo.

Over the course of the next five weeks, all the cells that have taken on these vital roles will keep dividing, folding, turning, and forming. By the time you're 10 weeks pregnant and probably looking forward to your first ultrasound, your baby will have a basic, functioning system of organs—a heart that beats, primitive kidneys that pass urine, and a brain that's beginning to send messages to other parts of the body.

